YASHWANTH KARUMANCHI

Machine Learning Engineer | GenAI, Agentic AI and LLMs, Computer Vision, Data Engineering & MLOps
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PROFESSIONAL SUMMARY

Machine Learning Engineer and M.S. Computer Science candidate (GPA 3.94) with experience building production-grade ML, computer vision, GenAI/RAG, MLOps, and large-scale data pipelines. Delivered measurable outcomes including 83% storage cost reduction, 4x faster retrieval, 39% RMSE reduction, 96.47% computer vision accuracy, and 5x faster processing.

EDUCATION

University of Utah, Salt Lake City, UT — M.S., Computer Science | GPA: 3.94/4.0 — Aug 2024 – May 2026
Coursework: Machine Learning, Deep Learning, Computer Vision, Data Mining, Advanced Algorithms, Large-Scale Data Processing

CVR College of Engineering, Hyderabad, India — B.Tech, Computer Science & Engineering — Dec 2020 – Jun 2024
Minor: Cybersecurity | GPA: 9.2/10

CERTIFICATIONS

AWS Certified Machine Learning Associate	2025–2028
AWS Certified Data Engineer Associate	2025–2028
Databricks Certified Machine Learning Associate	2026–2028
Databricks Certified Machine Learning Professional	2026–2028

TECHNICAL SKILLS

Programming	Python, SQL, Java, R, JavaScript, Bash
Machine Learning	Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Keras, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Feature Engineering, Imbalanced Learning
Deep Learning & Vision	CNNs, RNNs, Transformers, Vision Transformers, YOLO, DETR, Transfer Learning, Image Segmentation, Object Detection, OCR, Vision-Language Models
GenAI & LLMs	Agentic AI, AI Agents, Multi-Agent Systems, Tool Calling, RAG, LangChain, LlamaIndex, Hugging Face, Gemini, Claude, OpenAI APIs, Prompt Engineering, MCP, Embeddings, Vector Databases, Conversational Memory, LLM Evaluation, LoRA, QLoRA
Data Engineering	PySpark, Apache Spark, Kafka, Airflow, ETL/ELT Pipelines, Delta Lake, Parquet, Zarr, Large-Scale Data Processing, Data Quality Validation
MLOps	MLflow, Docker, Kubernetes, CI/CD, Model Deployment, Model Monitoring, Drift Detection, A/B Testing, SHAP, LIME, Reproducible ML Workflows
Cloud	AWS S3, EC2, SageMaker, Glue, Athena; GCP BigQuery, Vertex AI; Azure ML; Databricks
Databases	PostgreSQL, MySQL, MongoDB, Snowflake, Redis, NoSQL, Query Optimization
Visualization & Tools	Tableau, Power BI, Git, GitHub Actions, FastAPI, REST APIs, Streamlit, Linux

EXPERIENCE

Research Assistant --- University of Utah — **May 2026 – Present**

- Developing a semantic no-code data wrangling assistant that uses sentence embeddings and group similarity to recommend context-aware repairs for missing values, inconsistent categories, and fragmented tabular groups.
- Prototyped embedding-based and rule-based error analysis pipelines using SentenceTransformers, HDBSCAN, subgroup discovery, and decision trees to evaluate semantic explanations for data-quality issues.
- Designing semantic categorical canonicalization and repair recommendation workflows to merge equivalent values, suggest safer imputations, and generate plain-language explanations for visual data wrangling systems.

Machine Learning & Research Engineer — TraceAQ — **Jan 2026 – May 2026**

- Designed a cloud-native Zarr datastore for large-scale spatiotemporal air-quality modeling; optimized chunking, compression, and append workflows to reduce storage costs by 83% and accelerate retrieval by 4x.
- Built a semi-supervised PM2.5 prediction framework combining AirNow observations with low-cost sensor pseudo-labels, reducing RMSE by 39% and improving event detection accuracy by 41%.
- Constructed production-grade feature pipelines for GeoTIFF ingestion, S3 hourly workflows, meteorological features, wind-vector encoding, static rasters, and model-ready tensors for real-time inference.
- Improved model performance 66% over physics baseline and reduced false positives up to 71% through leakage-safe temporal splits, classifier-regressor gating, and threshold-aware evaluation.
- Established data-quality validation checks for temporal coverage, grid consistency, missing S3 keys, sensor availability, boundary-hour recovery, and pre-training statistical checks.

Teaching Assistant — University of Utah — **Aug 2025 – Dec 2025**

- Supported ML and computer vision coursework by guiding students through model training, debugging, preprocessing, evaluation metrics, and reproducibility best practices.
- Automated grading and evaluation workflows, increasing grading efficiency by 35%; resolved 200+ PyTorch, TensorFlow, package, GPU, and dataset-path issues.
- Standardized notebook execution, package setup, random seed control, and output verification, reducing environment-related failures by 30%.
- Reviewed classification, regression, CNN, object detection, and model evaluation implementations, providing feedback on correctness, performance, and reliability.

Research Assistant — University of Utah **Apr 2025 – Aug 2025**

- Delivered SINA, an end-to-end computer vision pipeline for schematic-to-netlist conversion using YOLO, connected-component logic, OCR, and vision-language models; achieved 96.47% accuracy.
- Surpassed prior approaches by 2.7x through improved component detection, wire reconstruction, symbol recognition, post-processing, and structured circuit mapping.
- Optimized deep learning architectures with kernel tuning, memory-efficient operations, and batching, reducing GPU usage by 60% and training time by 50%.
- Built validation and post-processing pipelines for components, OCR outputs, wire connections, and generated netlists, reducing detection errors by 25% and improving mapping accuracy by 12%.
- Contributed to research documentation, experiment tracking, result analysis, and manuscript preparation for AI-based schematic understanding.

Machine Learning Engineer — Watershed Support Services **Jul 2023 – Oct 2023**

- Reengineered a drone-based crop-health detection pipeline using modern computer vision architectures, augmentation, and class balancing, improving performance from 40% to 89% mAP.
- Processed and organized a 5,000-image agricultural dataset across 7 crop-health classes, improving label consistency, training quality, and downstream model reliability.
- Implemented scalable PySpark preprocessing pipelines 5x faster than local workflows; evaluated precision, recall, mAP, confusion matrices, and weak-class errors to improve detection accuracy.
- Delivered analytics outputs and dashboards for crop-monitoring insights, helping stakeholders interpret model predictions at field level.

PROJECTS

ARIA : AI CRM Agent *FastAPI, Gemini 3.1 Flash, Agentic AI, Google APIs, SSE, Google Sheets, JavaScript, Render*

- Built and deployed an agentic AI CRM platform with 50+ REST APIs, Google Sheets as a live database, and a 10-page frontend; designed an LLM-driven workflow engine that interprets user intent, invokes tools, and executes 13 CRM actions.
- Engineered tool-calling, conversational memory, Google Workspace integrations, AI lead scoring, and SSE-streamed imports supporting 5,000+ records; implemented prompt-injection defenses, validation, caching, and rate limiting, reducing API calls by ~60%.

NBA Conversational AI *FastAPI, Angular, RAG, Embeddings, ONNX*

- Engineered a full-stack conversational AI platform using FastAPI and Angular with RAG-based retrieval, embeddings, and vector search, improving retrieval accuracy by 14.7%.
- Reduced inference latency by 42% through ONNX INT8 quantization and optimized serving logic, supporting stable performance for 230 concurrent users.

Production MLOps Platform *MLflow, Docker, Drift Detection, SHAP, LIME*

- Architected a unified ML monitoring platform across classification, regression, NLP, and computer vision workloads with model tracking, data validation, and drift detection.
- Integrated KS-test, PSI, SHAP, LIME, and MLflow-based versioning to reduce root-cause analysis time by 40% and improve production deployment reliability.

HC18 Medical Image Segmentation *nnU-Net, MRI Segmentation, Deep Learning*

- Implemented nnU-Net based MRI segmentation workflows with optimized preprocessing, augmentation, and validation logic, achieving a Dice score of 0.74.
- Improved segmentation consistency by tuning training configurations, evaluating failure cases, and refining preprocessing for clinical image analysis workflows.

Distributed Data Processing Engine *PySpark, Data Cleaning, Feature Engineering*

- Designed a PySpark-based preprocessing system to handle imbalance, leakage, skewness, missing values, and outliers across large-scale tabular datasets.
- Processed 20GB of data in approximately 2 minutes and improved downstream model accuracy by up to 20% through cleaner training inputs.

Multi-Agent QA Generation System *LLMs, Agents, Tool-Based Workflows*

- Built a multi-agent LLM workflow for automated question-answer generation using tool-based planning, iterative refinement, and quality validation loops.
- Improved output consistency by separating generation, review, correction, and final formatting responsibilities across specialized agents.

PUBLICATIONS

Karumanchi Y., et al. “SINA: Circuit Schematic Image-to-Netlist Generator Using AI,” arXiv, 2026.
Karumanchi Y., et al. “Plantation Monitoring Using Drone Images: A Dataset and Performance Review,” arXiv, 2025.

LEADERSHIP & ACHIEVEMENTS

- Represented my state at the national level in India in Quad Speed Skating for 7 consecutive years, earning national-level appreciation, championships, and building resilience, consistency, and competitive focus.
- Led a multidisciplinary team at the Smart India Hackathon, earning second place overall while strengthening leadership, problem-solving, and execution under pressure.
- Served as Sergeant in the 1 TNCC Battalion, a cadet corps similar to U.S. cadet programs, holding the second-highest cadet rank and developing discipline, accountability, and leadership.
- Served as Captain of my CVR College Basketball Team across competitive seasons, developing team strategy, trust-building, communication, and performance under pressure, winning 3 championships during the time.
- Served as School President for an academic cohort of over 2,000 students, supporting student coordination, discipline, communication, and peer leadership.
- Awarded Individual Sports Championship for excellence across athletics, basketball, and skating, reflecting sustained hard work, adaptability, and pride in performance.